

PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

DAR ES SALAAM REGION

FORM FOUR MOCK EXAMINATION 2025

CHEMISTRY

GEPAM science hub

WhatsApp 0748978398

CODE:032/1

TIME:3:00Hours

Friday 18 July 2025 A.M

INSTRUCTIONS

1. This paper consists of section **A, B** and **C** with total of **eleven (11)** questions.
2. Answer **all** questions in section A and B and two (2) questions from section C.
3. Non – programmable calculators are allowed in the examination room.
4. Cellular phones and any unauthorized materials are **not** allowed in examination room.
5. Write your **Examination Number** on every page of your answer booklet(s)
6. The following constant may be used.
Atomic masses
 $H = 1, C = 12, Al = 27, Cl = 35.5, Mg = 24, S = 32, O = 16, Ca = 40$
Avogadro's number = 6.02×10^{23}
G.M.V at S.T.P = $22.4dm^3$
1 faraday = 96500 coulombs
Standard pressure = 760mmHg
Standard Temperature = 273 K
1 litre = $1dm^3 = 1000cm^3$

This paper consists of 4 printed pages

SECTION A (Carries 16 Marks)

Answer all questions from this section

1. For each of the following I – x choose the most correct answer from among the given alternatives and write its letter in the answer sheet provided.
- Desiccator is mainly used for
 - Gliding solid substances
 - Separating liquids
 - Dry solid substances
 - Storing reactive metals
 - Hydrating solid substances.
 - Which of the following chemicals is used to apply on burns?
 - Petroleum jelly
 - Gentian violet solution
 - Antibiotic solution
 - Soap
 - Ethanol
 - 17g of ammonia gas at S.T.P occupies the same volume as:
 - 1g of hydrogen gas
 - 16g of oxygen gas
 - 88g of carbon dioxide
 - 64g of Sulphur dioxide
 - 14g of nitrogen gas
 - A few substances were grouped in increasing order of their particles intermolecular forces of attraction. Using knowledge of kinetic theory of matter, which of the following is the correct order
 - Water, oxygen, chalk
 - Salt, juice, chalk
 - Nitrogen, carbon, water
 - Air, salt, oil
 - Hydrogen, dilute hydrochloric acid and stone
 - When colourless compound was heated a brown fumes of gas were evolved. The colourless gas is mostly likely to be from which of the following compound?
 - Ferrous sulphate
 - Silver chloride
 - Lead nitrate
 - Magnesium sulphate
 - Sodium hydroxide
 - Element found in a periodic table tends to show different behavior. When these elements lose or gain electrons, they tend to form
 - Ions
 - Allotropes
 - Molecules
 - Radical
 - Isotopes
 - A certain compound reacts with sodium to give hydrogen gas and it also reacts with acidified potassium dichromate to give a green solution. The compound is most likely to be
 - An alkane
 - An alcohol
 - Alkyne
 - A carboxylic acid
 - Alkene
 - How many moles of oxygen gas are required for the complete combustion of 4g of methane gas (CH_4) to form carbon dioxide and water
 - 0.05 moles
 - 0.50 moles
 - 0.25 moles
 - 0.025 mole
 - 0.225 moles

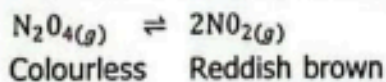
- ix. A solution is prepared by diluting 25cm^3 of 1.20M aluminium chloride solution to 250.00cm^3 . What is the molarity of the solution
- 0.2M
 - 0.12M
 - 6.50M
 - 1.20M
 - 0.10M
- x. One of the following is not an electrolyte
- Molten aluminium oxide
 - Concentrated sulphuric acid
 - Copper sulphate solution
 - Sodium chloride solution
 - Hard water.
2. Match the item in **List A** with the response in **List B** by writing the letter of the correct response beside the item number in the answer sheet provided

LIST A		LIST B	
i.	Polyester	A.	Used in electrical and thermal insulators
ii.	Polyvinyl cyanide	B.	Used as substitute for natural rubber
iii.	Perspex	C.	Making cloth items
iv.	Polyvinyl chloride	D.	Making cloth items and carpets
v.	Polythene	E.	Used to make water pipes, crates, electrical appliances and a substitute for leather
vi.	Neoprene	F.	Making packaging materials, basket, cups, bowls and a wide range of kitchen ware.
		G.	Used to treat water
		H.	Used to make table salt
		I.	Used as substitute for glass in furniture design, signal displays and lighting

SECTION B (54 Marks)

Answer **all** questions in this section

3. (a) (i) Explain how the separation of mixture so significance. (At least two points)
 (ii) Why are liquids and gases categorized as a fluids?
 (iii) What are the conditions for a substance to be called matter
- (b) A piece of potassium metal was put in a beaker containing cold water and piece of blue and red litmus papers.
 (i) Write a balanced chemical equation for the reaction
 (ii) List down any two changes observed during the course of the reaction
4. (a) Give the name and draw the safety signs which represented with the following chemical substance
 (i) Rat poison
 (ii) Ethanol
- (b) Briefly explain why it is not advised to use wood ashes for cleaning aluminium utensils
- (c) Give four (4) laboratory apparatuses that are made up of iron (steel) materials.
- (d) Give three differences when both solid ice and solid piece of paper are heated.
5. (a) Chemistry teacher prepare carbon dioxide by using 20g of grinded limestone by mixing with dilute nitric acid, he collected 200cm^3 of the gas. His students did the same experiments with 20g of ungrounded limestone where they collected 10cm^3 of the gas. With reasons explain the different in the volume of carbon dioxide collected.
- (b) The decomposition of gaseous dinitrogen tetroxide (N_2O_4) to form nitrogen dioxide (NO_2) is an endothermic process and can be represented by the following equilibrium?



- (i) Explain how could the pressure be adjusted so as to increase the intensity of reddish brown colour?
- (ii) Does the reaction cease when the mixture becomes reddish brown? Explain
- (iii) State and explain the colour change observed on cooling the reaction mixture
- (iv) Mention two ways reversing the change caused by cooling the reaction
6. (a) (i) Acid rain caused by the emission of acidic gases into the atmosphere from industries and vehicles. Briefly explain how acid – base neutralization can help to minimize this problem
- (ii) Using electron diagrams illustrate the formation of a molecule of ammonia gas.
- (iii) State two differences between a molecule of ammonia and sodium chloride
- (b) In neutralization reaction between sodium hydroxide and hydrochloric acid it forms compound Q and water only. What are the four (4) uses of compounds Q in our everyday lives.
7. (a) Metals are extracted from their ores. Using two important metals in the table below, identify the substance (s) added during extraction process and finally suggest their functions.
- | Name of the metal Extracted | Substances (s) added during extraction | Function of the substances |
|-----------------------------|--|----------------------------|
| Sodium | | |
| Iron | | |
- (b) Both sodium and iron are extracted by different methods. Explain reasons for this difference in their methods of extraction.
- (c) A gaseous compound consists of 86% carbon and 14% hydrogen by mass. At S.T.P. 3.2 dm³ of the compound has a mass of 6g. Calculate its
- (i) Empirical formula
- (ii) Molecular formula
8. (a) What is the relative molecular mass of a hypothetical gas Y if the mass of 560 cm³ of it measured at S.T.P. is 1.10g?
- (b) With the aid of balanced chemical equations explain how scum and lime are scale formed and state one disadvantage of each.
- (c) Compound A (C₄H₁₀O) is oxidized with acidified potassium dichromate to give compound B. When it is dehydrated with Conc. H₂SO₄, compound A gives compound C which undergoes hydrogenation to give compound D, that is capable of undergoing a substitution reaction. Identify the compounds B, C and D and write their structures.

SECTION C (30 Marks)

Answer only **two** questions from this section

9. (a) (i) State Faraday's first law of electrolysis
- (ii) Explain the three (3) industrial applications of electrolysis in our daily life
- (b) Dilute sulphuric acid was electrolyzed using carbon electrodes by passing a current of 0.2 amperes for half an hour.
- (i) Write down the electrode half reaction equations for the chemical reactions which occurred at both electrodes.
- (ii) Calculate the volume of the gas produced at each electrode
- (c) State two uses of each gas produced in the above electrolysis
- (d) How can you identify the presence of each gas produced at each electrode.
10. Explain the six (6) outcomes of the greenhouse effect.
11. Managing soil fertility is very crucial for farmers to have high yield season after season. If soil is not well managed loses its sufficient nutrients for proper plant growth and therefore plants and their yields are affected. Using six (6) points describe how nutrients are lost.

MARKING - GUIDE - 2025

CHEMISTRY 1

Q1

I	II	III	IV	V	VI	VII	VIII	IX	X
C	A	D	E	C	A	B	A	B	B

Q2

I	II	III	IV	V	VII
C	D	I	E	F	B

SECTION: B

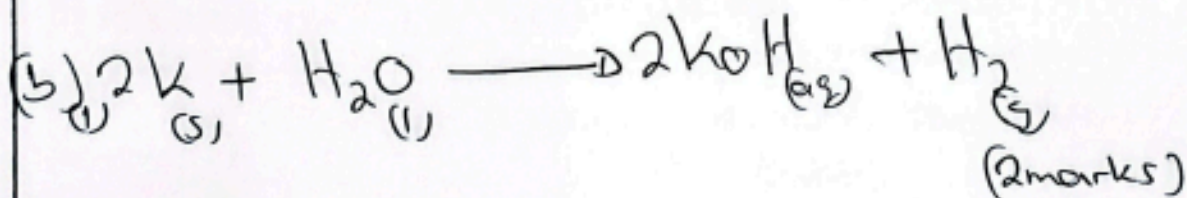
Q3

- @(i) - We get salt from sea water
 - We can extract useful oils from only seeds.
 - We can obtain metals from their ores.
 1mark@=2mark

(ii) Because their particles have ability to flow due to their constant random motion:
 (1mark).

~~(iii) Because their particles have ability to~~

(iii) - Mass
 - volume (space) (1mark)



(ii) - Evolution of colourless gas called hydrogen gas is observed

- The colour of red litmus paper turns blue since an alkaline solution (KOH) was formed while blue litmus paper will not change
 1mark@3marks

4 @ (1) Toxic (0.5 mark)



(1 mark)

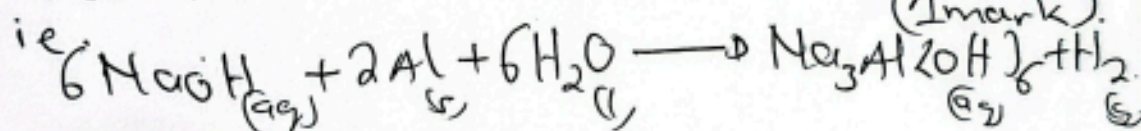


(0.5 mark)

Flammable (2 mark)

(b) Wood ashes are alkaline, thus if used to clean aluminium vessels would react and corrode the surface of the vessels.

(1 mark)



- (c) (i) Pair of tongs
(ii) Bunsen burner
(iii) Retort stand
(iv) Tripod stand
(v) Spatula.

0.5 @ any four (2 marks)

(d) Solid ice	Solid piece of paper
(i) No new substance is formed	New substance is formed.
(ii) Change is reversible	Change is irreversible
(iii) No energy is liberated	Energy is given out
(iv) There is ^{no} change of mass or weight	There is change of mass or weight.

1 mark @ = 3 marks
Any three points.

05 @ The grinded limestone have large surface area compared to ungrinded limestone allowing faster reaction with dilute nitric acid, hence producing greater volume of CO_2 while ungrinded limestone have small surface area allowing slower reaction with dilute nitric acid hence smaller volume of CO_2 . (2 marks)

(b)(i) The intensity of reddish brown colour is increased by decreasing pressure of the system because when pressure is decreased the equilibrium will shift towards the side with many moles of the gas. (2 marks)

(ii). Reaction doesn't cease because equilibrium will continue to shift back and forth b/w N_2O_4 and NO_2 ,
- The intensity of colour will change and reaction will not stop. (2 marks)

5(b)(iii) On cooling reaction, the colour will change from reddish brown to colourless, because reducing temperature will turn more production of H_2O_4 .
(2 marks)

(iv) Two ways of reversing the change caused by reaction.

(+) Increasing temperature

- Decreasing pressure

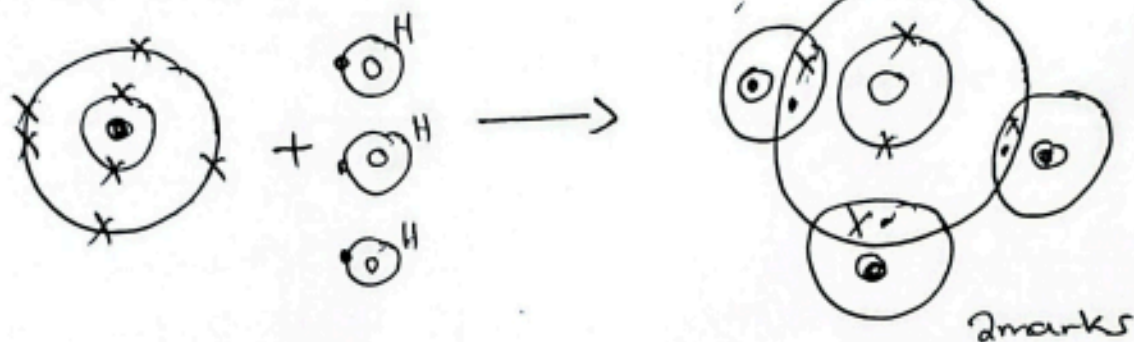
- Removing NO_2 / Increasing concentration of H_2O_4

or $\frac{1}{2}$ mark @ = 2 marks

06@

② (i) The acidic gases should be neutralized before being released into the atmosphere. This can be achieved by placing or fitting basic substances such as Ca(OH)_2 in exhaust pipes and chimney so as to trap and react with acidic gases.
(1 mark)

(ii) Ammonia molecule: NH_3
 $\text{N} : 2 : 5$ $\text{H} : 1$



(iii) Difference between NaCl and NH_3

NaCl	NH_3
Formed by transfer of electrons	Formed by sharing of electrons.
Solid at room temperature	Liquid/gases at room temperature
Conduct heat and electricity	Does not conduct heat and electricity.

Any two points
 1 mark @ = 2 marks

6b) Uses of compound (Salt).

- Control of soil pH.
- Used as antacids
- As inorganic fertilisers
- Alleviation of health disorders
- Production of pesticides.

Any four
1 mark @ 4 marks.

7a) Name of metal extracted	Substance added	Function of the Substance.
Sodium.	calcium chloride 0.5 marks	To lower melting point of sodium chloride 0.5 marks
Iron.	Coke 0.5 marks	Acts as reducing agent 0.5 marks

6b) Both sodium and iron are extracted by different methods, because of their position in the reactivity series, sodium is very reactive metal and extracted through electrolysis and iron is less reactive metal extracted through reduction with carbon.
(1 mark)

7(c) Elements	C	H	1/2 mark
Percentage composition	86%	14%	1/2 mark
Divide Percentage composition by its R.A.M.	$\frac{86}{12}$ $= 7.1667$	$\frac{14}{1}$ $= 14$	1/2 mark
Divide by smallest number	$\frac{7.1667}{7.1667}$ $= 1$	$\frac{14}{7.1667}$ $= 1.953$	1/2 mark
Approximate to whole number	1	2	1/2 @ 2 1/2 marks

∴ The empirical formula of a compound is CH_2 .

(1/2 mark)

To find R.M.M from the given data.

- Volume of gaseous compound = 3.2 dm^3 .
- Volume at s.t.p = 22.4 dm^3
- Mass of gaseous compound = 6 g
- Molar mass = ?

from

$$\text{Amount of substance} = \frac{\text{Mass of gas}}{\text{Molar mass}} = \frac{\text{Volume of gas}}{\text{molar volume}}$$

1/2 mark

$$\frac{\text{Mass of gas}}{\text{Molar mass}} = \frac{\text{Volume of gas}}{\text{Molar volume of gas.}}$$

$$\frac{6 \text{ g}}{M_r} = \frac{3.2 \text{ dm}^3}{22.4 \text{ dm}^3}$$

$$M_r = \frac{6 \text{ g} \times 22.4 \text{ dm}^3}{3.2 \text{ dm}^3}$$

1/2 mark

7(c) cont - - -

$$M_r = 42 \text{ g/mol.}$$

$$\therefore \text{R.M.M} = 42 \text{ ~~at~~ } \text{ (0.5 marks)}$$

Recall

$$(\text{Emp. f}) \times n = \text{M.f}$$

but

$$n = \frac{\text{R.M.M}}{\text{E.F.M}}$$

$$n = \frac{42}{14} \text{ ~~at~~ }$$

$$n = 3$$

$$(\text{CH}_2)_n = \text{M.f}$$

$$(\text{CH}_2)_3 = \text{M.f} \text{ (0.5 marks)}$$

$$\text{C}_3\text{H}_6 = \text{M.f.}$$

$$\therefore \text{Molecular formula} = \text{C}_3\text{H}_6 \text{ (1 mark)}$$

50

$$\text{Mass} = 1.10\text{g}$$

$$\text{Volume} = 560\text{cm}^3 = 0.56\text{dm}^3$$

$$\text{Molar mass of (Y)} = ?$$

(each marks)

from.

$$n = \frac{\text{Volume}}{\text{molar volume}}$$

$$n = \frac{0.56\text{dm}^3}{22.4\text{dm}^3}$$

$$= 0.025\text{mol.} \quad (\text{each marks})$$

but again

$$n = \frac{\text{mass}}{\text{molar mass}}$$

$$\text{Molar mass} = \frac{\text{mass}}{n}$$

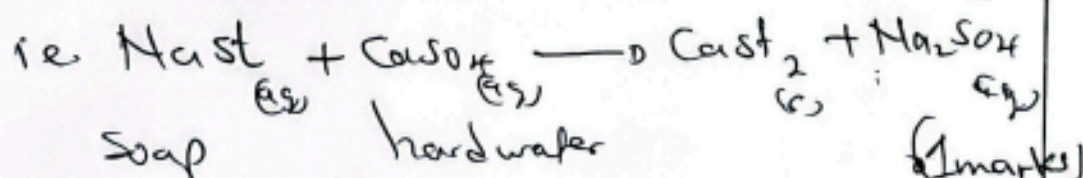
$$= \frac{1.10\text{g}}{0.025\text{mol}}$$

$$= 44\text{g/mol}$$

$$= \text{H}_2\text{CO}_3/\text{mol}$$

∴ Relative molecular mass of Y is $\text{H}_2\text{CO}_3/\text{mol}$ (1 marks)

⑧ - Scum is a white insoluble substance formed when Soap reacts with hard water (each marks)



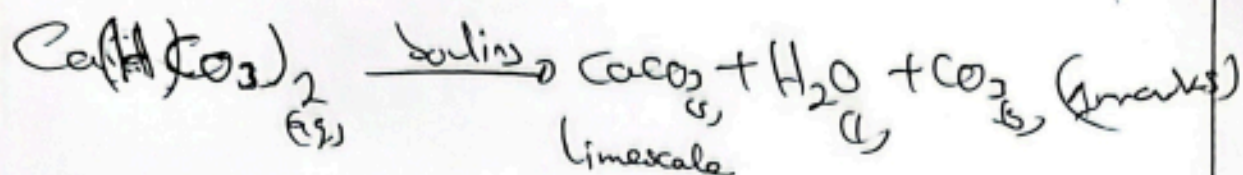
(1 marks)

Dis-advantage

- Causes wastage of soap in cleaning and washing process

(each marks)

- 8 (b) Limescale is a coat or layer deposited on the wall of the water pipes or boilers and it is formed as a result of chemical decomposition of $\text{Ca}(\text{HCO}_3)_2$ or $\text{Mg}(\text{HCO}_3)_2$ in hard water to form insoluble CaCO_3 or MgCO_3



Disadvantages

Reduces efficiency of electrical appliances such as heaters: (2 marks)

- (c) Compound B: - Butanoic acid
- $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$ (1 mark)

C - Butan-1-ene
- $\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2$ (1 mark)

D - Butane
- $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$ (1 mark)

SECTION C. (30 MARKS)

9(a)(i) Faraday's first law state that:

"The mass of an element discharged at the electrode during electrolysis is directly proportional to the quantity of electricity passed through the electrolyte"

(01 marks)

(ii) - Production of gases

- Purification of Copper (metals)

- Electroplating

01 @ = (03 marks)

(b) (i) At Cathode: $2H_{(aq)}^{+} + 2e^{-} \rightarrow H_{2(g)}$ (01 marks)

At anode: $4OH_{(aq)}^{-} \rightarrow 2H_{2O(l)} + O_{2(g)} + 4e^{-}$

(01 marks)

(ii) Given

$$I = 0.2A, \quad t = 30 \times 60 \\ = 1800 \text{ sec}$$

$$\text{from } Q = It \\ 0.2 \times 1800$$

$$\therefore Q = 360 \text{ Coulombs} \quad (01 \text{ mark})$$

$$\therefore \begin{array}{r} 1F \Rightarrow 96500C \\ ? \Rightarrow 360C \end{array}$$

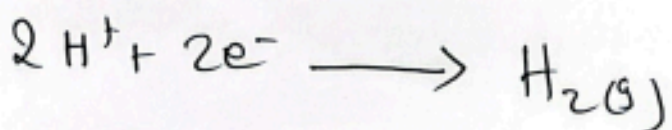
$$\frac{1 \times 360}{96500}$$

9.

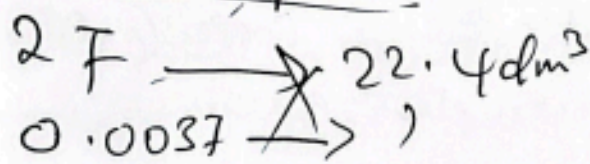
$$\frac{1 \times 360}{96500}$$

$$= 0.0037 \text{ Faradays} \quad (0.1 \text{ marks})$$

At Cathode



from equation



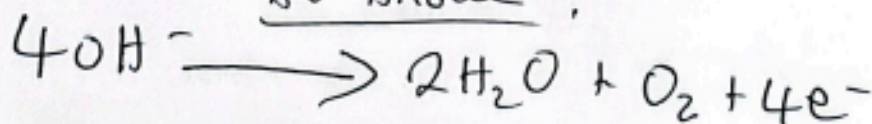
$$\frac{0.0037 \times 22.4}{2}$$

$$= 0.04 \text{ dm}^3$$

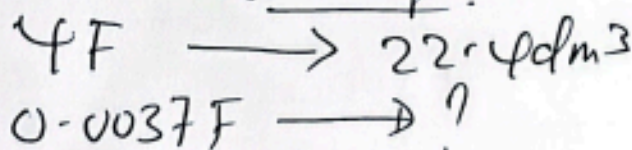
∴ 0.04 dm³ of H₂ were produced at Cathode

(0.2 marks)

At Anode



from eqn.



$$\frac{0.0037 \times 22.4}{4} = 0.02 \text{ dm}^3$$

∴ 0.02 dm³ of O₂ were produced at anode. --- (02 marks)

(C)

Uses of Hydrogen gas

- used as a fuel in rocket
 - used in hardening of oil
 - used as Oxy-hydrogen flame in welding
- any two : 00½ @ = (01 marks)

Uses of Oxygen gas

- used by deep-sea divers
- Support Combustion
- used to support life of patients in ICU and ~~parent~~ premature babies in incubators.

00½ @ = (01 marks)

(d)

- Hydrogen gas tested by producing a pop sound. (00½ marks)
 - Oxygen gas it relights a glowing wooden splint
- any two (00½ marks)

10 Introduction - 2 marks

Main body

Outcomes of greenhouse effects

(i) Global warming.

(ii) Increased drought

(iii) Rising of oceans / sea level

(iv) Loss of biodiversity

(v) Increased health risk

(vi) Food shortage

(vii) More severe storms 2 marks @ 2 marks

Conclusion - (1 mark)

Point - 0.5 mark
Exp. 1 mark

11 Introduction - 2 marks

Main body

• Loss of soil nutrients

(i) Monocropping

(ii) Soil erosion

(iii) Overgrazing

(iv) Deforestation

(v) Leaching

(vi) Burning of vegetation 0.2 marks @ 2 marks

Conclusion - (1 mark)